

ECO 202 — Practice Exam 2 — Version A

Worked solutions

Fall 2026 | Classes 6–9

Equivalent exact expressions and valid alternative reasoning are acceptable. For practice study, attempt the paper first, then compare reasoning, close these solutions, and reconstruct any missed method independently.

Multiple-choice key

Question	Answer	Reason and distractor audit
1	D	The target is a numerical feature of all city residents. A is a sample count; B is a respondent statistic; C describes the response process, not population transit use.
2	B	Pair formation uses baseline information and treatment is randomized within pairs. A uses chosen treatment and outcomes; C concerns sampling, not assignment; D is deterministic assignment.
3	A	Additivity gives $\mathbb{P}(\Omega) = \mathbb{P}(\Omega) + \mathbb{P}(\emptyset)$, hence the empty event has probability zero. B confuses the empty event with the full space; C and D contradict this model identity.
4	A	Disjointness gives zero intersection, whereas independence would require the strictly positive product. B forgets the positive union; C confuses the definitions; D is false because the conditioning probabilities are positive.

5. A workforce survey

(a) The unit is a person who worked for the firm. The target contains all 100 people who worked there at any time last year; the frame is the 80 current employees; the selected sample is the 40 randomly selected current employees; the responding sample is the 20 who answer.

(b) The parameter is the complete target-population mean

$$\mu = \frac{80(5) + 20(15)}{100} = 7 \text{ hours.}$$

The respondent sample statistic is $\bar{y} = 120/20 = 6$ hours.

(c) Excluding former employees produces undercoverage, also described here as survivorship selection. The excluded group has the higher mean, so this frame restriction lowers the mean from 7 to 5. Differential nonresponse favors longer-overtime current employees and tends to raise the respondent mean relative to the current-employee frame. These mechanisms act in opposite directions at different stages. The observed respondent mean is nevertheless 6, below the target mean 7 in the supplied complete records. This final comparison follows from those numbers; the names of competing mechanisms alone would not establish their net size.

(d) Even an accurate census of all 80 current employees yields their mean, 5 hours. The former employees remain excluded, so the desired mean across all 100 is not recovered. The proposal

removes sampling and nonresponse errors within the current roster under the stated complete-response assumption, but it does not fix undercoverage of the target.

6. Blocking a training experiment

- (a) This is randomized block assignment, with low and high prior earnings as pre-treatment blocks. Treatment is the training offer; control is no offer. Blocking assigns treatments among already enrolled people. Stratified sampling instead selects people from a population into a study.
- (b) Each block supplies six equally likely pairs, with independent selections, so there are $6 \times 6 = 36$ equally likely offer assignments, each of probability $1/36$. Offering all four low-earnings people is possible under completely randomized selection of four from eight, with probability $1/70$, but is impossible under this block design.
- (c) Equal cell sizes give offered mean $(8 + 12)/2 = 10$ and control mean $(5 + 11)/2 = 8$. The difference is 2 thousand dollars. It estimates the average causal effect of the offer under the experiment's implementation; chance assignment variation means a realized difference need not equal the fixed average effect for the eight units. Volunteer recruitment also does not establish generalization to all workers.
- (d) Differential outcome-related attrition selects the responding groups after random assignment and can destroy the original comparability. A difference among remaining respondents need not identify the original offer effect. Useful responses include collecting outcomes for all assigned units through administrative records or intensive follow-up regardless of assignment. A promise to “control for variables” does not by itself solve missing outcomes related to their unobserved values.

7. A small sample space

- (a) The six equally likely unordered samples are

$$\Omega = \{AB, AC, AD, BC, BD, CD\},$$

each of probability $1/6$. The events are

$$U = \{AB, AC, AD, BC, BD\}, \quad V = \{AB, AD, BD\}.$$

- (b) $\mathbb{P}(U) = 5/6$, $\mathbb{P}(V) = 3/6 = 1/2$, and $U \cap V = V$, giving $\mathbb{P}(U \cap V) = 1/2$. Since $(5/6)(1/2) = 5/12 \neq 1/2$, the events are not independent. Learning that C is absent ensures that at least one of A or B is selected.
- (c) Conditioning on A leaves AB, AC, AD , equally likely. Only AB includes B, so the answer is $1/3$. B's unconditional inclusion probability is $1/2$; conditioning on A uses one of the two sample slots and leaves only one of the three remaining people to be selected.
- (d) The exact probability C is absent is $1/2$ under the specified SRS mechanism. A finite simulation can yield a frequency $53/100$ rather than $1/2$ because its realized samples fluctuate. This discrepancy alone neither disproves the model probability nor proves an implementation error. The simulation frequency is a random approximation under the mechanism, not a redefinition of its exact probability.

8. A card with a hidden side

- (a) Label cards 1–6 and label each physical side a or b. The 12 elementary outcomes are $(1, a), (1, b), \dots, (6, a), (6, b)$, each of probability $(1/6)(1/2) = 1/12$. The two RR cards supply four red-revealing outcomes and the three BR cards supply three more, for seven total. Three outcomes both use a mixed card and reveal red.
- (b) $\mathbb{P}(R) = 7/12$, $\mathbb{P}(M) = 3/6 = 1/2$, and $\mathbb{P}(R \cap M) = 3/12 = 1/4$. Independence would require $(7/12)(1/2) = 7/24$, which differs from $1/4 = 6/24$. Therefore they are not independent. Equivalently, $\mathbb{P}(R | M) = 1/2 \neq 7/12$.

- (c) Among the seven equally likely red-revealing outcomes, four come from RR cards and have a red hidden side. Thus

$$\mathbb{P}(\text{hidden red} \mid \text{revealed red}) = \frac{4}{7}.$$

Bayes gives the same result: $\mathbb{P}(\text{RR} \mid R) = [(2/6)(1)]/(7/12) = 4/7$.

- (d) The five compatible cards are not equally likely after observing red. Each RR card reveals red with probability 1; each BR card does so with probability $1/2$. Therefore each RR card has twice the conditional probability of each BR card (respectively $2/7$ and $1/7$). The BB card becomes impossible. The probability of a blue hidden side is $3/7$, coming from BR cards, and $4/7 + 3/7 = 1$.